Categories, Proofs, and Processes: Assignment 5

Wira Azmoon Ahmad

26 Feb 2024

Question 1

(a)

The set-indexed product of a set of objects $\{A_i\}_{i\in I}$ is the Cartesian product

$$\prod_{i\in I} A_i = \left\{ g : I \to \bigcup_{i\in I} A_i \middle| \forall i \in I : g(i) \in A_i \right\}$$

with its arrows $\pi_j(g) = g(j)$ for all $j \in I$.

Let C be an object with a set of arrows $f_i : C \to A_i$. Then the unique arrow $\langle f_i\rangle_{i\in I}$ is defined where for all $c \in C$,

$$\langle f_i\rangle_{i\in I}(c) := i \mapsto f_i(c)$$

For each $j \in I$, and for all $c \in C$,

$$\begin{aligned} \pi_j \circ \langle f_i\rangle_{i\in I}(c) &= \pi_j(i \mapsto f_i(c)) \\ &= (i \mapsto f_i(c))(j) \\ &= f_j(c) \end{aligned} \tag{1}$$

so $\forall j \in I, \pi_j \circ \langle f_i\rangle_{i\in I} = f_j$.

$$\begin{tikzcd}
 & \prod_i A_i \arrow[dl, "\dots"'] \arrow[d, dashed, "\pi_j"] \arrow[dr, "\dots"] & & \\
\dots & A_j & \dots & \\
 & & & C \arrow[ulll, "\dots"] \arrow[ull, "f_i"] \arrow[ul, "\dots"'] \arrow[uull, dashed, bend right, "\langle f_i\rangle_i"']
\end{tikzcd}$$

Suppose there is another arrow $h : C \to \prod_{i\in I} A_i$ such that $\forall j \in I, \pi_j \circ h = f_j$. Then $\forall c \in C$,

$$\begin{aligned} \pi_j \circ h(c) &= \pi_j(h(c)) \\ &= h(c)(j) \\ h(c)(j) &= f_j(c) \\ \implies h(c) &:: j \mapsto f_j(c) \end{aligned} \tag{2}$$

So $h = \langle f_i\rangle_{i\in I}$ and the arrow is unique.

(b)

When $|I| = 2$, we get binary products. The product of 2 objects A_i, A_j for $i, j \in I$ is the object $A_i \times A_j$ with arrows $A_i \xleftarrow{\pi_i} A_i \times A_j \xrightarrow{\pi_j} A_j$ such that for any object C and pair of arrows $A_i \xleftarrow{f_i} C \xrightarrow{f_j} A_j$, there is a unique arrow $\langle f_i, f_j \rangle : C \to A_i \times A_j$ such that $\pi_i \circ \langle f_i, f_j \rangle = f_i$ and $\pi_j \circ \langle f_i, f_j \rangle = f_j$.

When $|I| = 0$, we get terminal objects. The nullary product for $I = \emptyset$ is the terminal object T such that for any object C, the arrow $\tau_C : C \to T$, is the unique arrow from C to T.

(c)

Let (L, γ) be the limit of $D : \mathcal{I} \to \mathcal{C}$. Then the arrows are defined as

$$\{\gamma_J : L \ \to D\ J\}_{J \in \mathrm{Ob}(\mathcal{I})}$$

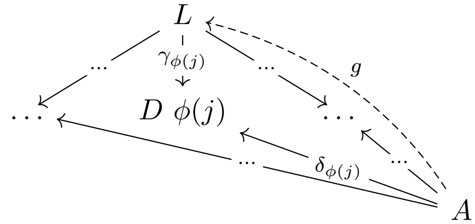
where the only arrows in $\mathcal{J}$ are identities 1_J so $D\ 1_J = 1_{D\ J}$. Since (L, γ) is the terminal object in **Cone**(D), for every other cone to $D, (A, \delta)$, there is a unique arrow $g : A \to L$ such that $\forall J \in \mathrm{Ob}(\mathcal{I}), \gamma_J \circ g = \delta_J$.

$$\begin{tikzcd}
 & D\ J \\
A \arrow[ur, "\delta_J"] \arrow[r, "g"'] & L \arrow[u, "\gamma_J"']
\end{tikzcd}$$

By using the functor $S : \mathbf{Set} \to \mathbf{Cat}$ from Assignment 3 Exercise 1, let $S\ I = \mathcal{I}$, so there's an isomorphism $\phi : I \cong \mathrm{Ob}(\mathcal{I})$. Then set $\forall j \in I$,

- $D\ \phi(j) := A_j$
- $L := \prod_{j \in I} A_j$
- $\gamma_{\phi(j)} := \pi_j$
- $A := C$
- $\delta_{\phi(j)} := f_j$
- $g := \langle f_j \rangle_{j \in I}$

so the limit of the diagram matches the defintion of the set-indexed products.



(d)

Proposition 1.1. *If $\mathcal{D}$ has all set-indexed products, then so too does the functor category $[\mathcal{C}, \mathcal{D}]$.*

Proof. Suppose $\mathcal{D}$ has all set-indexed products.

Let I be the set of indices, and $\{F_i : \mathcal{C} \to \mathcal{D}\}_{i \in I}$ be a set of functors. First, we construct the product functor $\prod_{i \in I} F_i : \mathcal{C} \to \mathcal{D}$ and prove its functoriality.

$$\prod_{i \in I} F_i(C) := \prod_{i \in I} (F_i\ C)$$

$$\prod_{i \in I} F_i(f : C \to C') := \prod_{i \in I} (f_i : F_i\ C \to F_i\ C') := \langle f_i \circ \pi_i \rangle_{i \in I}$$

where $\pi_i : \prod_{i \in I} F_i \to F_i\ C$, which exists in $\mathcal{D}$.

Let $C \xrightarrow{f} C' \xrightarrow{g} C''$ be arrows.

$$\begin{aligned}
\prod_{i \in I} F_i(g \circ f) &= \langle (g \circ f)_i \circ \pi_i \rangle_{i \in I} \\
&= \prod_{i \in I} (g \circ f)_i \circ \langle \pi_i \rangle_{i \in I} \\
&= \prod_{i \in I} g_i \circ \prod_{i \in I} f_i \circ \langle \pi_i \rangle_{i \in I} \\
&= \langle g_i \circ \pi_i \rangle_{i \in I} \circ \prod_{i \in I} f_i \circ \langle \pi_i \rangle_{i \in I} \\
&= \langle g_i \circ \pi_i \rangle_{i \in I} \circ \langle f_i \circ \pi_i \rangle_{i \in I} \\
&= \prod_{i \in I} F_i(g) \circ \prod_{i \in I} F_i(f)
\end{aligned} \tag{3}$$

$$\begin{aligned}
\prod_{i \in I} F_i(1_C) &= \langle 1_{F_i\ C} \circ \pi_i \rangle_{i \in I} \\
&= \langle \pi_i \rangle_{i \in I} \\
&= \prod_{i \in I} 1_{F_i\ C} \\
&= 1_{\prod_{i \in I} (F_i\ C)} \\
&= 1_{\prod_{i \in I} F_i(C)}
\end{aligned} \tag{4}$$

So the product functor satisfies functoriality.

Next, we define and show the naturality of the functor projections. For an arbitrary natural transformation $t_j : \prod_i F_i \to F_j$, define $\forall C$ in $\mathcal{C}$,

$$t_{j,C} = \pi_{j,C} : \prod_i F_i\ C \to F_j\ C$$

which exists as a projection in $\mathcal{D}$. Let $f : C \to C'$ be an arrow in $\mathcal{C}$, and $f_j : F_j\ C \to F_j\ C'$ be an arrow in $\mathcal{D}$. Then by the property of the product, we have

$$f_j \circ \pi_{j,C} = \pi_{j,C'} \circ \langle f_i \circ \pi_i \rangle_i$$

So naturality is satisfied.

$$
\begin{tikzcd}
G \arrow[d, dashed, "\langle s_i \rangle_i"] \arrow[dd, bend right=60, "s_j"'] \\
\prod_i F_i \arrow[d, "t"] \\
F_j
\end{tikzcd}
\qquad\qquad
\begin{tikzcd}
G\ C \arrow[r, "G\ f"] \arrow[d, dashed, "\langle s_{i,C} \rangle_i"] \arrow[dd, bend right=60, "s_{j,C}"'] & G\ C' \arrow[d, dashed, "\langle s_{i,C} \rangle_i"] \arrow[dd, bend left=60, "s_{j,C'}"] \\
\prod_i F_i\ C \arrow[r, "\langle f_i \circ \pi_i \rangle_i"] \arrow[d, dashed, "\pi_{j,C}"] & \prod_i F_i\ C' \arrow[d, dashed, "\pi_{j,C'}"] \\
F_j\ C \arrow[r, "f_j"'] & F_j\ C'
\end{tikzcd}
$$

Let again $f : C \to C'$ be an arrow in $\mathcal{C}$, Suppose we have some functor G and a set of natural transformations $s_i : G \to F_i$. Define the natural transformation $\langle s_i \rangle_i : G \to \prod_i F_i$ such that

$$\langle s_i \rangle_{i,C} = \langle s_{i,C} \rangle_i : G\ C \to \prod_i F_i\ C$$

By naturality of s_i, we have

$$f_i \circ s_{i,C} = s_{i,C'} \circ G\ f$$

From the property of the product, we have that

$$\pi_{j,C} \circ \langle s_i \rangle_{i,C} = s_{j,C}, \quad \pi_{j,C'} \circ \langle s_i \rangle_{i,C'} = s_{j,C'}$$

Then these imply that

$$f_j \circ \pi_{j,C} \circ \langle s_i \rangle_{i,C} = \pi'_j \circ \langle s_i \rangle_{i,C'} \circ G\ f$$
$$\pi_{j,C'} \circ \langle f_i \circ \pi_i \rangle_i \circ \langle s_i \rangle_{i,C} = \pi_{j,C'} \circ \langle s_i \rangle_{i,C'} \circ G\ f$$

Since this is true $\forall j \in I$, then by the property of the product, there is the unique arrow

$$\langle f_i \circ \pi_i \rangle_i \circ \langle s_i \rangle_{i,C} = \langle s_i \rangle_{i,C'} \circ G\ f : G\ C \to \prod_i F_i\ C'$$

So the naturality of $\langle s_i \rangle_i$ is satisfied. Since $\forall C$ in $\mathcal{C}$, and $\forall j \in I$, we have some $\pi_{j,C}$ that

$$\pi_{j,C} \circ \langle s_i \rangle_{i,C} = s_{j,C}$$

where $t_j = \pi_{j,C}$, then

$$\forall j \in I : t_j \circ \langle s_i \rangle_i = s_j$$

Now, we show the uniqueness of the natural transformation. Suppose there is some other natural transformation $u : G \to \prod_i F_i$ such that $\forall j \in I, t_j \circ u = s_j$. Then for all C in $\mathcal{C}$, $\forall j \in I, \pi_j \circ u_C = s_{j,C}$ for some π_j. Then there is a set of mappings

$$\pi_j \circ u_C = s_{j,C} : G\ C \to F_j\ C$$

By property of the product, there is a unique mapping

$$\langle \pi_i \circ u_C \rangle_i = \langle s_{i,C} \rangle_i : G\ C \to \prod_i F_i\ C$$

such that $\forall j \in I, \pi_j \circ \langle \pi_i \circ u_C \rangle_i = s_{j,C}$. Then for all C in $\mathcal{C}$,

$$\begin{aligned} s_{j,C} &= \pi_j \circ \langle \pi_i \circ u_C \rangle_i \\ &= \pi_j \circ \langle \pi_i \rangle_i \circ u_C \\ &= \pi_j \circ 1_{\prod_i F_i \ C} \circ u_C \\ &= \pi_j \circ u_C \end{aligned} \tag{5}$$

By property of the product, there is the unique arrow

$$\langle s_{i,C} \rangle_i = \langle \pi_i \circ u_C \rangle_i = u_C : G\ C \to \prod_i F_i\ C$$

So $\forall C$ in $\mathcal{C}, u_C = \langle s_{i,C} \rangle_i$. Thus, $\langle s_i \rangle_i : G \to \prod_i F_i$ is the unique natural transformation that satisfies the properties defined above. $\square$

(e)

Proposition 1.2. *The Yoneda embedding $Y : \mathcal{C} \to [\mathcal{C}^{op}, \boldsymbol{Set}]$ preserves set-indexed products, when they exist.*

Proof. Suppose $\mathcal{C}$ has set-indexed products. Let $\{C_i\}_i$ be a set of objects in $\mathcal{C}$ with the product, object $\prod_i C_i$, with arrows $\pi_j : \prod_i C_i \to C_j$. We show that for the set of objects $\{Y\ C_i\}_i := \mathcal{C}(-, C_i)_i$ in $[\mathcal{C}^{op}, \mathbf{Set}]$, the object with arrows

$$Y\ \prod_i C_i := \mathcal{C}(-, \prod_i C_i)$$

$$\forall j \in I, Y\ \pi_j := \mathcal{C}(-, \pi_j) : \mathcal{C}(-, \prod_i C_i) \to \mathcal{C}(-, C_j)$$

is their set-indexed product.

By theorem from lectures, Y is full and faithful. Hence, there is an bijection between sets of arrows

$$\phi_{A,B} : \mathcal{C}(A, B) \cong [\mathcal{C}^{op}, \mathbf{Set}](Y\ A, Y\ B)$$

Then by the property of the product, since for any object C' and set of arrows $f_i : C' \to C_i$ in $\mathcal{C}$, there is the unique arrow $\langle f_i \rangle_i : C' \to \prod_i C_i$ such that $\forall j \in I, \pi_j \circ \langle f_i \rangle_i = f_j$. For the corresponding objects $Y\ C', \{Y\ C_i\}_i, Y\ \prod_i C_i$, with the set of arrows $\{Y\ f_i\}_i = \{\phi_{C',C_i}(f_i)\}_i$ in $[\mathcal{C}^{op}, \mathbf{Set}]$, there must be the unique arrow

$$Y\ \langle f_i \rangle_i = (\phi_{C',\prod_i C_i})(\langle f_i \rangle_i) : Y\ C' \to Y\ \prod_i C_i$$

such that $\forall j \in I, (Y\ \pi_j) \circ (Y\ \langle f_i \rangle_i) = (Y\ f_j)$. The embedding preserves set-indexed products when they exist.

$\square$

Question 2

(a)

First, define the functor

$$\Delta : [\mathcal{P}^{op}, \mathbf{Set}] \to [\mathcal{P}^{op}, \mathcal{B}]$$

$$\forall C \in \mathrm{Ob}(\mathcal{P}), \Delta(F\ C) = \begin{cases} \emptyset & \text{if } F\ C = \emptyset \\ \{*\} & \text{if } F\ C \neq \emptyset \end{cases}$$

$$\forall C \in \mathrm{Ob}(\mathcal{P}), \Delta(f_C : F\ C \to G\ C) = \begin{cases} 1_\emptyset & \text{if } F\ C = G\ C = \emptyset \\ \emptyset \to \{*\} & \text{if } F\ C = \emptyset \wedge G\ C \neq \emptyset \\ 1_{\{*\}} & \text{if } F\ C \neq \emptyset \end{cases}$$

Functoriality is satisfied since $\forall C \in \mathrm{Ob}(\mathcal{P})$,

$$\begin{aligned} \Delta(((g : G \to H) \circ (f : F \to G))_C) &= \Delta((g_C : G\ C \to H\ C) \circ (f_C : F\ C \to G\ C)) \\ &= \begin{cases} 1_\emptyset & \text{if } F\ C = G\ C = H\ C = \emptyset \\ \emptyset \to \{*\} & \text{if } F\ C = \emptyset \wedge H\ C \neq \emptyset \\ 1_{\{*\}} & \text{if } F\ C \neq \emptyset \end{cases} \\ &= \begin{cases} 1_\emptyset \circ 1_\emptyset & \text{if } F\ C = G\ C = \emptyset \wedge G\ C = H\ C = \emptyset \\ (\emptyset \to \{*\}) \circ 1_\emptyset & \text{if } F\ C = \emptyset \wedge (G\ C = \emptyset \wedge H\ C \neq \emptyset) \\ 1_{\{*\}} \circ (\emptyset \to \{*\}) & \text{if } (F\ C = \emptyset \wedge G\ C \neq \emptyset) \wedge H\ C \neq \emptyset \\ 1_{\{*\}} \circ 1_{\{*\}} & \text{if } F\ C \neq \emptyset \wedge G\ C \neq \emptyset \end{cases} \\ &= \Delta(g_C : G\ C \to H\ C) \circ \Delta(f_C : F\ C \to G\ C) \end{aligned} \tag{6}$$

$$\Delta(1_A) = \left\{ \begin{matrix} 1_\emptyset & \text{if } A = \emptyset \\ 1_{\{*\}} & \text{if } A \neq \emptyset \end{matrix} \right\} = 1_{\Delta A}$$

By definition of Δ, there is a bijection on each arrow map $\Delta_{F,G}$ for each pair of objects F, G in $[\mathcal{P}^{op}, \mathbf{Set}]$, since all cases are covered, where $\forall A, C \in \mathrm{Ob}(\mathcal{P}), \mathcal{P}(A, C) \in \{\emptyset, \{*\}\}$, and there is only a single mapping for each arrow f, where $\forall C \in \mathrm{Ob}(\mathcal{P})$,

$$f_C : \begin{cases} 1_\emptyset \\ \emptyset \to \{\leq\} \\ 1_{\{\leq\}} \end{cases} \iff \Delta(f_C) = \begin{cases} 1_\emptyset \\ \emptyset \to \{*\} \\ 1_{\{*\}} \end{cases}$$

Thus there is a bijection on arrows

$$\forall A, C \in \mathrm{Ob}(\mathcal{P}), \Delta_{A,C} : [\mathcal{P}^{op}, \mathbf{Set}](A, C) \cong [\mathcal{P}^{op}, \mathcal{B}](\Delta A, \Delta C)$$

Since the Yoneda embedding is full and faithful as proven in lecture, combining this with the above bijection on arrows using Δ, there is a bijection on arrows for Y' using $\Delta \circ Y : \mathcal{P} \to [\mathcal{P}^{op}, \mathcal{B}]$. So Y' is full and faithful.

(b)

From lecture,

$$\hat{S} : \mathcal{P}^{op} \to \mathbf{Set} :: \begin{cases} \hat{S}(A) = 1 & \text{if } A \in S \\ \hat{S}(A) = 0 & \text{if } A \notin S \end{cases}$$

We have that a downward-closed set S is such that

$$A \in S \wedge B \leq A \implies B \in S$$

so define $S_B := B^{\downarrow} := \{A | A \leq B\}$ which satisfies the condition.

Then we can see that

$$\forall B \in \mathrm{Ob}(\mathcal{P}), \{A | \hat{S}_B(A) = 1\} := \{A | \mathcal{P}(A, B) = \{*\}\} = \{A | A \leq B\} := S_B$$

Then any downward-closed set S_B has a corresponding $\hat{S}_B$, so there is a 1-to-1 correspondence.

Representable functors are naturally isomorphic to the covariant Hom-functor $\mathcal{P}(-, A)$ for some A, which has the correspondence to the set

$$\{B | \mathcal{P}(B, A) = \{*\}\} = \{A | A \leq B\} := A^{\downarrow}$$

(c)

Each functor $\hat{S}, \hat{T}$ has a corresponding downward-closed set S, T in $\mathcal{P}$. The transformation $\hat{S} \to \hat{T}$ then has the corresponding subset relationship $S \subseteq T$, which is a unique arrow in the preorder of downward-closed sets with binary relation $\subseteq$.

(d)

If $\hat{S} \cong \hat{T}$, then $S \subseteq T \wedge T \subseteq S$, so $S = T$. Thus $\hat{S} = \hat{T}$ from the 1-to-1 correspondence.

(e)

By (a), the Yoneda embedding $Y : \mathbb{Q} \to [\mathbb{Q}^{op}, \mathbf{Set}]$ restricts to a full and faithful functor $Y' : \mathbb{Q} \to [\mathbb{Q}^{op}, \mathcal{B}]$.

By (b), objects in $[\mathbb{Q}^{op}, \mathcal{B}]$ are in 1-to-1 correspondence with downward-closed sets in $\mathbb{Q}$.

Using Dedekind pre-cuts, each $r \in \mathbb{R}^{\infty}$ is uniquely represented by a downward-closed set. Then each $r \in \mathbb{R}^{\infty}$ is in 1-to-1 correspondence with objects in $[\mathbb{Q}^{op}, \mathcal{B}]$.

We show that the arrow mapping of $[\mathbb{Q}^{op}, \mathcal{B}]$ and $\mathbb{R}^{\infty}$ is bijective.

If $r_1, r_2 \in \mathbb{R}^{\infty}$ such that $r_1 \leq r_2$, then there are downward-closed sets S_{r_1}, S_{r_2} such that $S_{r_1} \subseteq S_{r_2}$, for surjectivity.

By (c), there is only one arrow between objects S_a, S_b in $[\mathbb{Q}^{op}, \mathcal{B}]$, which is mapped to the corresponding inequality $a \leq b$ in $\mathbb{R}^{\infty}$, for injectivity.

$[\mathbb{Q}^{op}, \mathcal{B}]$ fully and faithfully embeds $\mathbb{R}^{\infty}$.

(f)

Since $[\mathbb{Q}^{op}, \mathcal{B}]$ is a partial order, products of objects are the greatest lower bound. Using $\mathbb{Q}$ as the set of indices, and the set of objects $\{F_i : \mathbb{Q}^{op} \to \mathcal{B}\}_{i \in \mathbb{Q}}$, the product is defined,

$$\forall q \in \mathbb{Q}, \prod_{i \in \mathbb{Q}} F_i(q) := q^{\downarrow}$$

which uniquely represents a real number $r \in \mathbb{R}, r > q$ where $r^{\downarrow} = q^{\downarrow}$. Since set-indexed products are limits of diagrams from 1(c), every real number is a limit of rational numbers.